Executive Briefing & Facility Vulnerability Report

CRITICAL SECURITY AUDIT FINDING: 100% UNREGISTERED / UNKNOWN SUBJECTS
Zero facial profiles match an enrolled employee database. **All 1,652 detected personas (SUBJ-01 to SUBJ-1652) represent UNKNOWN PERSONS (Pending Identification).** Tiers reflect sighting frequency only—NOT confirmed employment or authorized clearance.

1. Executive Mission Overview

An autonomous forensic and biometric surveillance audit was conducted over 30 continuous calendar days of native 3K CCTV footage covering the RO Water Treatment Plant rooftop. The audit established continuous tracking of all human activity in the mechanical wing, measured shift-by-shift recurrence patterns, resolved historical video clipping blindspots, and logged every direct physical contact event against mission-critical equipment assets.

2. Key Operational Metrics

Metric Category	Audit Result	Operational Security Context
Total Facility Coverage	**713.66 Recorded Hours**	30 continuous calendar days (100% of Sept 2026 footage)
Total Discovered Personas	**1,652 Unique Personas**	100% Unknown / Unregistered Individuals grouped via ArcFace clustering
High-Frequency Unknowns	**12 Individuals**	Tier 1 recurring unknown subjects (50+ detections each)
Direct Equipment Contacts	**18 Unknown Individuals**	Verified touching multiport valves, skid controls, or electrical breaker switch
Pristine Incident Clips	**179 Events (8.20 GB)**	Lossless H.265 stream cuts with +/-60s pre/post buffers
Processed Face Vectors	**36,226 Embeddings**	512-D ArcFace vectors passing strict quality threshold (Q >= 40)

3. Operational Shift Workload Distribution

Shift A (Morning, 06:00 – 13:59): 24,625 detections (68.0%) — Peak operating window where recurring unknown subjects routinely operated valves.
Shift B (Afternoon, 14:00 – 21:59): 11,188 detections (30.9%) — Secondary window with multiple unknown subjects traversing the corridor.
Shift C (Night Vigilance, 22:00 – 05:59): 413 detections (1.1%) — Night visits, perimeter walkthroughs, and main breaker switch contacts.

4. Incident Investigations & Resolutions

1. **Day 1 Backwash Valve Incident (09:53:38):** Re-extracted full continuous 4-minute clip (Day01_0953_Manual_Incident_Audit.mp4) and cataloged unknown person as **SUBJ-1652**.
2. **Day 1 Night Power Breaker Contact (22:27:36):** Re-extracted seamless 6-minute clip (Day01_2227_Manual_Incident_Audit.mp4) spanning the 22:28 file split.
3. **Sorting Column Inversion Fix:** Rebuilt all 30 clipping indices with strict wall-clock sorting, eliminating timestamp inversions across all 179 incident events.

5. Immediate Security Recommendations

1. Mandatory Personnel Enrollment: Conduct immediate formal identity verification with facility HR to associate names and employment status with all 1,652 unknown subject clusters.
2. Physical Access Restriction: Secure Zone 1A/B Valves and Zone 2 Power Switch with physical lockout/tagout (LOTO) padlock boxes.
3. Real-Time Spatial Alerting: Implement automated audible alerts when any unauthorized centroid breaches the 40px buffer of Zone 1 or Zone 2.